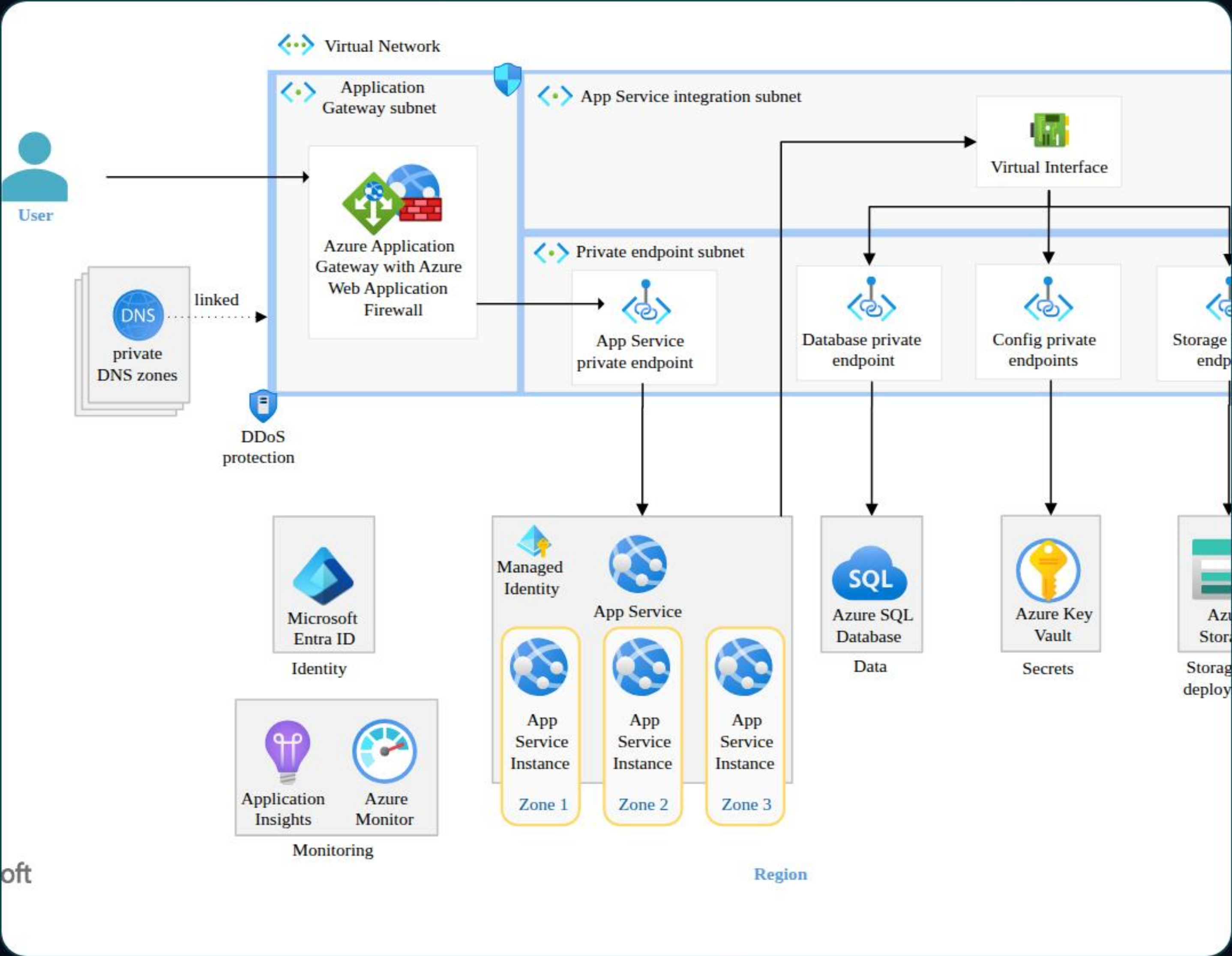


SECURE GITOPS GATEKEEPER

KubeGuard AI

Azure-Based AI-Powered Kubernetes Deployment Risk Analysis Platform

-  Azure Secured
-  Terraform IaC
-  AI Risk Model



PRESENTER NOTES:

Welcome the Azure Review Panel and Cloud Architects. Introduce KubeGuard AI. Emphasize that this is an enterprise-grade admission control and deployment validation layer engineered specifically to address runtime failure models in highly secured Azure Kubernetes clusters.

01. Executive Summary & Core Value

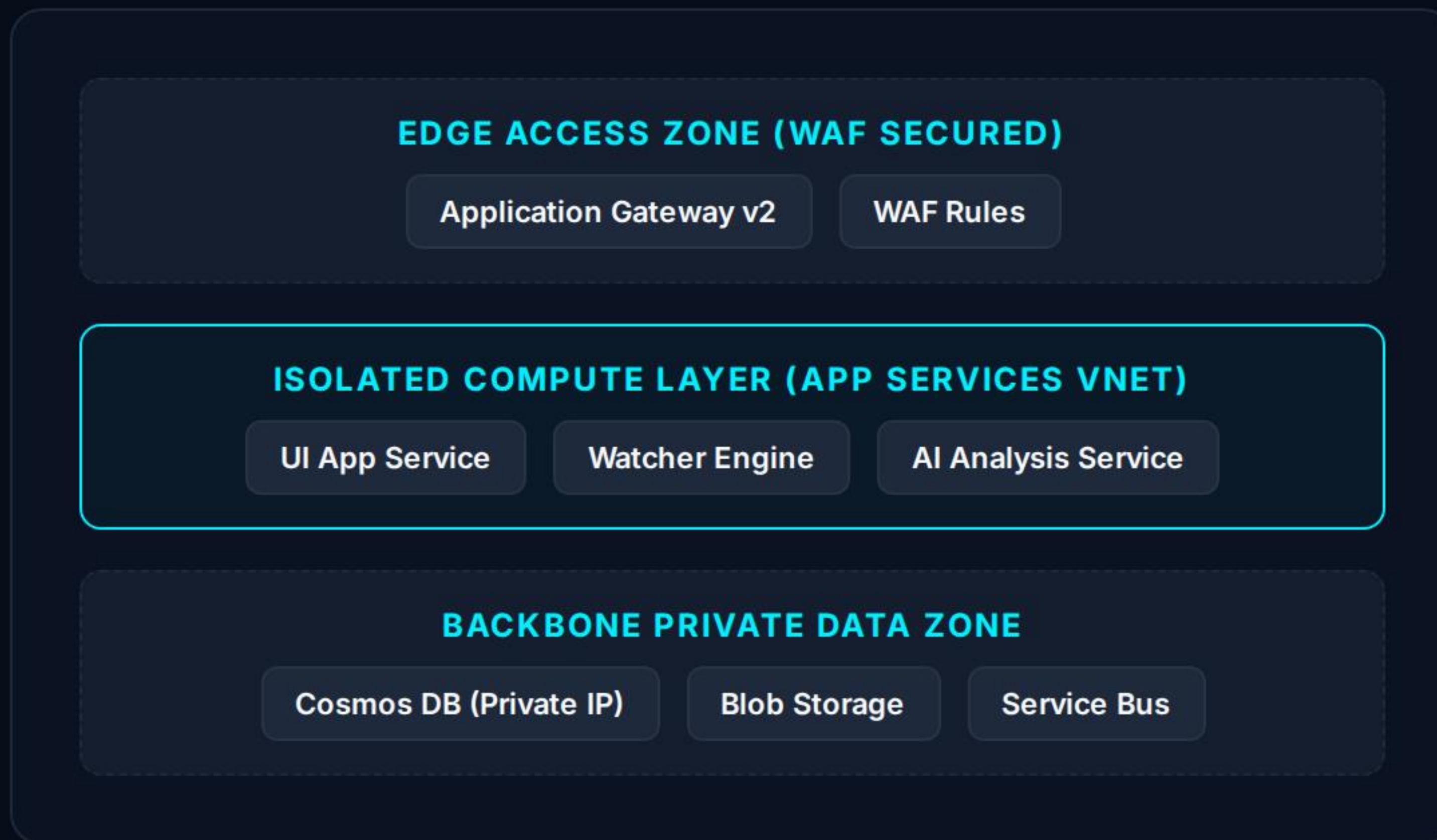
⚠️ The Governance Gap

- **Outage Risks:** Complex Kubernetes configurations routinely pass basic YAML schema linter gates but fail structurally during live rolling updates.
- **Context Blindness:** Current CI/CD tools lack dynamic access to Prometheus telemetry, executing manifests blindly against degraded cluster environments.
- **GitOps Automation:** ArgoCD pipelines bypass security authorization frameworks during dynamic automated synchronizations.

✅ KubeGuard AI Solution

- **Pre-Deployment Gate:** Enforces an intelligent, mathematically verified predictive validation gate before Kubernetes workloads scale.
- **ML-Powered Insights:** Analyzes cluster telemetry, node resource usage trends, and past historical configuration anomalies to grade deployment risk.
- **Human-in-the-Loop:** Intercepts dynamic pipelines and prompts designated DevOps leads for contextual sign-off on medium-to-high risk tasks.

02. Enterprise Azure Reference Architecture



Architectural Highlights

- **Strict Perimeter Defense:** Public inbound calls terminate securely at Azure Application Gateway. Direct microservice exposure is strictly blocked.
- **Virtual Network Injection:** Azure App Services are seamlessly linked with dedicated subnets via Regional VNet Integration for private outbound traffic routing.
- **Cosmos DB Private Link:** Relies fully on Private Endpoints within the VNet. All analytical databases have direct public network routing completely disabled.

03. Infrastructure Service Matrix

Azure Service	Structural Responsibility	Key Features Leveraged	Security Baseline
 Entra ID	Identity Provider & Governance	Unified SSO, RBAC Custom Claims	Conditional Access Enabled
 App Service	Microservice Compute Hosting	Regional VNet Outbound Integration	Managed System Identity
 App Gateway	Layer-7 Application Load Balancing	WAF Engine OWASP Rules enforcement	TLS 1.3 Strict Termination
 Cosmos DB	NoSQL Configuration Database	Private Endpoint Connectivity Only	RBAC Key-less Authentication
 Blob Storage	Analytical Risk Report Store	Secure Container Networking Filters	Customer-Managed Key Encryption
 Service Bus	Inter-service Message Broker	Event Routing Topics & Queues	SasToken Authorization

PRESENTER NOTES: Explain that we run a completely modern cloud architecture with passwordless connections. System Managed Identities are used for App Services to access Storage, Database, and Key Vault, avoiding credentials leakage.

04. Network Boundary & DNS Isolation

VNet-1: Edge

Dedicated Application Gateway Subnet.
Enforces inbound security rules via Azure
Network Security Groups (NSGs).

VNet-2: Compute

Hosts UI and Analysis engines. Interconnects
with Data layers via VNet Peering over high-
speed Azure Backbone.

VNet-3: Storage

No public interface. Hosts Private Endpoints
mapping storage nodes directly to private non-
routable VNet IPs.

CORE PRIVATE DNS ZONES IMPLEMENTED

`privatelink.azurewebsites.net`

`privatelink.mongo.cosmos.azure.com`

`privatelink.blob.core.windows.net`

05. Unified Authentication & RBAC

JWT Claims Verification Flow

- 1. User Authentication:** Sign-on initialized against Entra ID directory.
- 2. Claims Processing:** Identity Token issued with embedded organization and workspace roles.
- 3. Gateway Enforcement:** Edge processes JWT, validates public key, and asserts programmatic route access.

Admin Role

Highest authorization authority inside KubeGuard AI control panels.

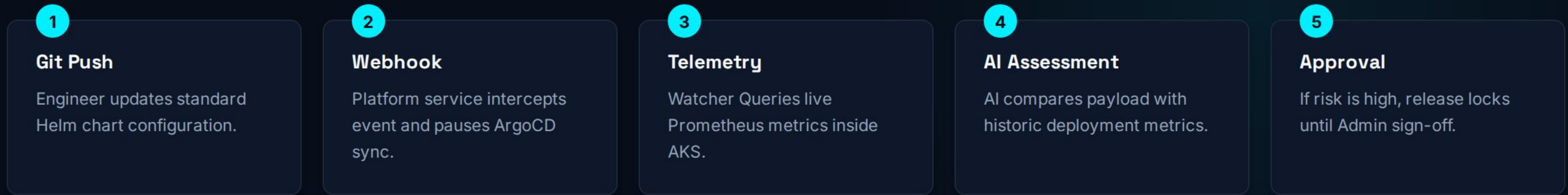
- Override automated rollout pauses
- Audit platform security settings
- Global project creation

DevOps Lead

Primary operational staff for day-to-day continuous deployment monitoring.

- Publish deployment metadata
- Inspect machine-learning risk scores
- Cannot force bypass blockades

06. End-to-End GitOps Validation Flow



Result: Zero blind deployments. Failures are caught proactively before workloads propagate into critical zones.

07. IaC Governance & Risk Scoring Engine

Terraform-Based Architecture

The entire Azure network, security parameters, DNS records, and microservice resources are defined and provisioned deterministically using modular Terraform HCL blueprints.

- **Automated Env Replication:** Replicates dev, staging, and production clusters natively inside independent VNets with zero manual portal touch points.
- **Immutable State Auditing:** Prevents unauthorized manual state drifting, aligning cloud networks directly with verified Git check-ins.

ANALYTICAL PREDICTIVE RISK ENGINE

$$R_d = \sum_{i=1}^N W_i \times C_i + \alpha \cdot \theta_{\text{telemetry}}$$

Risk Index R_d evaluates structural configurations (C) scaled against weights (W) and combined with real-time cluster telemetry dynamics (θ).

Image Sources



<https://learn.microsoft.com/en-us/azure/architecture/icons/images/baseline-app-service-architecture.svg>

Source: learn.microsoft.com